

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES

First Assessment 2081

Subject: Research Methodology

Course No: MDS 601

Level: MDS / II Year / III Semester

Full marks: 45

Pass Marks: 22.5

Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [5 × 3 = 15]

1. What are the different types of hypothesis? What is the main function of null hypothesis?
2. Explain the characteristics of scientific method.
3. Explain the difference between research method and Research methodology.
4. Define literature survey. What are the purposes of literature survey?
5. What are the steps involved in the research design process?

Group B [5 × 6 = 30]

6. What is scientific research? Describe the major steps involved in the scientific research process?
7. Define research problem. What steps are to be followed in identifying a research problem?
8. Explain the difference between a concept and a variable.
9. Explain the difference between fundamental research and applied research with suitable examples.

OR

What is case study research? Describe limitations and characteristics of a case study research.

10. Prepare a bibliographic entry in APA and IEEE method for the following :
- a) Book with one author
 - b) Journal article with two author

OR

A manager notices the number of grievances in the organization is increasing. The manager wishes to investigate this occurrence. What research design seems appropriate for the study? Give appropriate explanation.

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First Assessment 2081

Subject: **Advanced Data Mining**
Course No: **MDS 602**
Level: **MDS / II Year / III Semester**

Full marks: **45**
Pass Marks: **22.5**
Time: **2hrs**

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5 × 3 = 15]

1. What are the different types of the warehouse? Explain with their uses.
2. Define KDD. What are the steps involved with KDD workflow?
3. What is association analysis?
4. Define data scaling and normalization with examples.
5. Write a K-NN algorithm with an example.

Group B [5 × 6 = 30]

6. Explain how classification can be used for data mining? Explain with examples.

OR

What is the Apriori Algorithm? Discuss with its use case.

7. List and explain the uses of the different types of data visualization techniques

OR

Write ID3 Algorithm. With an example, discuss the pros and cons of the decision tree based classification technique.

8. What is data pre-processing? List the data pre-processing techniques useful for numeric data
9. In the context of data mining, how should a data scientist decide whether to use supervised or unsupervised learning for a given problem? Discuss the key factors that influence this decision and provide examples of scenarios where each type of learning would be appropriate.
10. How does ANN work? Explain the working of the perceptron neuron for classification tasks.

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First Assessment 2081

Subject: Technique for Big Data
Course No: MDS 603
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5 × 3 = 15]

1. What do you mean by big data? Briefly point out the challenges of implementing big data. 2.5
2. What is Hadoop Streaming? What is it used for? 2.5
3. Write the key features of Apache Hadoop. 2.5
4. How does HDFS works? 2
5. What are the components of Spark Shell? What are they used for? 2

Group B [5 × 6 = 30]

6. Explain the characteristics of big data. 5
7. Explain the operations of RDD. Distinguish between RDD, DataFrame and Dataset. 5

OR

What is Structured Streaming? Explain the working mechanism of Structured Streaming? 5

8. Explain the various configuration of Hadoop along with their characteristics. 5
9. Explain the replica placement and risk-aware policy in HDFS.

OR

Explain the architecture of HDFS with block diagram 5

10. Explain the key components of Hadoop. 5

$$25 + 11 = 36$$

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First Assessment 2081

Subject: Regression Analysis
Course No: MDS 605
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Explain cross sectional data with suitable example.
2. Show that mean of error/residual is always zero.
3. How do you measure the goodness of fit in multiple regression analysis. Discuss two model selection criteria briefly.
4. Describe the null and alternative hypothesis and state the test statistic and decision criteria to test the significance of individual regression coefficient (β_i).
5. Suppose VIF for three independent variables X_1 , X_2 , and X_3 are 3.18, 4.21, and 13.18 respectively then what decision do you make for the multicollinearity?

Group B [5x6= 30]

6. Describe the general regression model and also describe the assumptions of the regression model.

OR

Show that $\hat{\beta} = (X'X)^{-1}(X'Y)$

7. A state fisheries commission wants to estimate the number of bass caught in a given lake during a season in order to restock the lake with the appropriate number of young fish. The commission could get a fairly accurate assessment of the seasonal catch by extensive "netting sweeps" of the lake before and after a season, but this technique is much too expensive to be done routinely. Therefore, the commission samples a number of lakes and records Y (the seasonal catch, thousands of bass per square mile of lake area), X_1 (the number of lakeshore residences per square mile of lake area), X_2 (the size of lake in square miles), X_3 ($X_3=1$, if the lake has public access, 0 if not) and X_4 (a structure index. (Structures are weed beds sunken trees, drop-offs and other living places for bass.) The commissions run the data in SPSS to fit the regression model and the incomplete output is obtained as follows:

Predictor	Unstandardized Coefficient		t-value	p-value
	Coefficient	Standard error		
Constant	-1.9378	0.9081	?	
Residence (X_1)	0.0193	0.0101	?	0.560
Size (X_2)	0.3323	0.2458	?	0.120
Access (X_3)	0.8355	0.2250	?	0.001
Structure (X_4)	0.0477	0.0050	?	0.001

Analysis of variance (ANOVA) table:

Source	Df	SS	MSS	F	p-value
Regression	?	21.04	?	?	0.000
Error	?	2.82	?		
Total	19	23.86			

- Complete above coefficient and ANOVA table and describe the best-fitting regression equation for these data.
 - Test the significant impact of X_3 (Access) on Y (the seasonal catch, thousands of bass per square mile of lake area) at 5% level of significance.
 - Compute the standard error of the estimate and interpret its meaning.
 - What percentage of the variation in gross earnings is explained by three variables?
 - At the 0.05 level of significance, is the regression model significant as a whole?
- State the unbiasedness property of estimator of slope/regression coefficient.
 - Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.

OR

Explain the consequences of autocorrelation on unbiasedness, bestness and consistent properties of OLS.

- Explain Breusch Pagan Godfrey test to examine the presence of heteroskedasticity in data.

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First Assessment 2081

Subject: Monte Carlo Methods

Course No.: MDS 607

Level: MDS /II Year / III Semester

Full Marks: 45

Pass Marks: 22.5

Time: 2 hrs

Candidates are required to give their answers in their own word as far as practicable
Attempt All Questions.

Group A [5 x 3 =15]

1. Explain the meaning of "Random numbers". 2
2. Elucidate the significance of "Random variables". 2
3. Clarify the meaning of "Continuous state space". 2
4. State the limiting theorem. Also explain its significance. 1.5
5. How do you estimate errors in Monte Carlo methods? Describe. 2

Group B [5×6=30]

6. Discuss the significance of Monte Carlo methods in "Data Science".

OR

Discuss Monte Carlo method to estimate value of π . 4.5

7. Distinguish Simple and importance sampling in Monte Carlo methods of Integration

OR

How do you evaluate

$$I = \int_{-1}^1 dx \int_{-1}^1 dy g(x, y)$$

with

$$g(x, y) = \exp(-x^2 - y^2 - (x - y)^2/2)$$

Using Monte Carlo. Illustrate with algorithm.

8. Explore the significance of "Reversible Markov chain". 4
9. Discuss the term "Equilibrium distribution" with examples. Explain an algorithm to get "Equilibrium distribution" using Python code. 4
10. Explain the meaning of "Markov Chain". Also explore its role in Monte Carlo methods. 4

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Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [5x3=15]

1. Point out the difference between research methodology and research methods.
2. Explain the difference between reference and bibliography.
3. Explain nominal scale and interval scale with suitable examples.
4. Describe the APA method of citation for single author and multiple author with example.
5. Point out the characteristics and purpose of descriptive research design.

Group B [5x6=30]

6. What is the difference between a review of literature and a summary of literature?
7. Describe the major steps involved in the scientific research process?
8. Explain the concept and types of reliability.
9. What is research report? What purpose does it serve? Why is reporting of research study important?

OR

Explain the contents of research proposal. Is there any difference between a funded research proposal and an academic research proposal?

10. The mean systolic blood pressure of a certain group of people was found to be 125 mm of Hg with standard deviation of 15 mm of Hg. Calculate sample size to verify the result at 5 % level of significance if error do not exceed 2. Also find sample size if sample is selected from population of size 500.

OR

Distinguish between sampling and Non sampling error.

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Attempt All Questions.

Group A [5x3=15]

- ~~1.~~ How to build a data warehouse? Explain in the context of retail business?
2. Explain the concept of "slice" and "dice" in OLAP. Provide an example scenario to illustrate how these operations can be used for data analysis.
- ~~3.~~ Discuss the descriptive data mining techniques.
- ~~4.~~ What are the differences between Normalization vs Standardization in numerical data?
- ~~5.~~ List out the advantages and disadvantages of the K-NN algorithm.

Group B [5x6=30]

6. What is Clustering? Differentiate between K-Means and DBScan algorithm.

OR

What is classification? Discuss any decision tree induction technique with the help of an example.

7. Explain exploratory data analysis techniques discussing the context of data mining.

OR

What is the kernel function in SVM? Explain Gaussian Kernel Radial Basis Function (RBF) with examples.

8. Suppose you have to solve the multi-class classification problem. After research you concluded that none of the single ML models are capable of classifying all labels correctly. Explain how you deal with this situation?
9. A financial institution, ABC Bank, wants to identify fraudulent transactions within its vast dataset of customer transactions from the core banking system. They decide to use clustering techniques for outlier detection. Explain what an outlier is in this context and describe how ABC Bank can use clustering to detect these outliers, providing examples to illustrate the process.
10. Company XYZ, an online retail platform, wants to improve its product recommendation system to increase sales and enhance customer satisfaction. They are considering implementing an association-based recommendation system. Describe how Company XYZ can build this system, explaining the steps involved and providing examples of how the recommendations would work in practice.

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Subject: Techniques for Big Data
Course No: MDS 603
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Full marks: 45
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Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What are the defining characteristics of big data?
2. When is it appropriate to use the Hadoop Federation configuration?
3. What types of failures does MapReduce address?
4. What is the function of ZooKeeper in HBase?
5. What are the main features of RDD Transformations?

Group B [5x6=30]

6. What does big data mean? Discuss the current trends and challenges surrounding big data.
7. Explain Apache Hadoop and its main components.
8. What is MapReduce? Describe the process of executing a MapReduce job on a cluster.

OR

Describe HDFS, its working process, and its architecture.

9. Compare and contrast NoSQL and relational databases.

OR

Explain the read and write operations in HBase.

10. Describe the architecture of Hive and its key components.

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Group A [5x3=15]

1. Describe BLUE properties briefly.
2. What is multicollinearity? Explain variance inflation factor criteria briefly.
3. Describe forward selection procedure briefly.
4. Explain techniques for checking normality.
5. How do you measure the goodness of fit in logistic regression.

Group B [5x6= 30]

6. Describe the general regression model and also describe the assumptions of the regression model.

OR

A sample of 34 stores in a chain is selected for a test-market study of Pokhara Noodles. All the stores selected have approximately the same monthly sales volume. Measurements are obtained by taking price and promotional expenditure as independent variables and sales as a dependent variable. The partial output obtained from SPSS Software is as following.

ANOVA

Model	Sum of square	df	Mean square	F	p-Value
Regression	39472731	?	?	?	0.000
Residual	?	?	?		
Total	52093678	?			

Coefficients

Model	Unstandardized Coefficient		T	p-value
	Bi	Standard error		
Constant	5837.521	628.150	?	
Price	-53.217	6.852	?	0.000
Promotion	3.613	0.685	?	0.000

- a. Test the significance effect of promotion on sales at 1% level of significance.
- b. What percentage of total variation in sales is explained by price and promotional expenditure?
- c. Compute adjusted coefficient of determination and interpret its meaning.
- d. Compute standard error of estimate and interpret its meaning.
- e. Test the overall fit of the model at 5% level of significance.
- f. Develop the regression model and predict the sales when price Rs.70 and promotional expenditure is Rs.300.

7. Suppose the residuals for a set of data collected over 10 consecutive time period are as follows. Let the model consists of two independent variables

Time	1	2	3	4	5	6	7	8	9	10
Residuals	-3	-4	-6	-2	-1	0	2	3	4	7

Compute the Durbin Watson test statistic and test for significance at .level 05. Is there evidence of positive autocorrelation among residuals?

8. Describe Breusch-Godfrey Lagrange Multiplier Test (LM test) to test the autocorrelation.
9. Explain the impact of Heteroskedasticity on BLUE property. Explain Breusch-Pagan-Godfrey Test briefly.
10. When do you use the logistic regression? Explain logit model briefly.

OR

Define dummy variable. How does interchanging the code numbers affect the y-intercept and regression coefficient (slope) when we consider dummy variable as an independent variable.

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Attempt All Questions.

Group A [5x3=15]

1. What is Bayesian inference? Explain.
2. Elucidate the meaning of "prior distribution".
3. Clarify the idea "burn-in period".
4. Explain motivating vignettes of Statistical inferences.
5. Describe likelihood principle.

Group B [5x6=30]

6. State and prove Metropolis algorithm in MCMC.

OR

Discuss the main idea of Gibbs Sampling.

7. Consider bivariate normal distribution. Discuss necessary theory and Algorithm to solve this problem using Gibbs sampling.

OR

Explain algorithm to apply Gibbs Sampling to study Hierarchical model.

8. Discuss the significance of "reparametrization" in MCMC.

9. Explore convergence diagnostic in various MCMC algorithm.

10. Ultrasound tests done near the end of the first trimester of a pregnancy are often used to predict the sex of the baby. However, the errors made by radiologists in reading ultrasound results are not symmetric, in the following sense: girls are virtually always correctly identified as girls, while boys are sometimes misidentified as girls (in cases where the gender organ is not clearly visible, perhaps due to the child's position in the womb). More specifically, a leading radiologist states that $P(\text{test} + | G) = 1$ and $P(\text{test} + | B) = 0.25$ where "test +" denotes that the ultrasound test predicts the child is a girl. Thus we have 25% false positive rate for girl, but no false negatives. Suppose a particular woman's test comes back positive for girl, and we wish to know the probability she is actually carrying a girl. Assuming 48% of babies are girls, we can use Bayes Rules where "boy" and "girl" provide the $J = 2$ mutually exclusive and exhaustive cases. Calculate $P(G|+)$.
